

Exam in Statistical Physics and Solid State Physics

Exam date: 15.1.2022

Exam time: 4 hours.

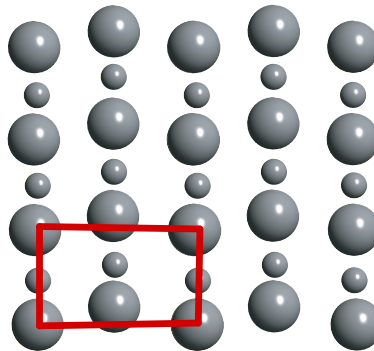
Permitted help: books, notes. No electronics are permitted (phones, calculators, computers.....).

This exam consists of two parts: The solid state physics part contains 4 pages and the statistical physics part contains 3 pages.

Answers can be given in English or Danish. For the questions in which you have to supplement figures, you can hand in these original pages instead of re-drawing the figures.

1 Crystal structure (2 points)

Figure 1 shows a two-dimensional lattice of atoms. Draw the smallest possible unit cell into the figure and state how many atoms there are in this unit cell.



*2 small atoms
2 large atoms
in unit cell*

Figure 1: A two-dimensional lattice of atoms.

2 Reciprocal lattice (2 points)

Figure 2 shows the reciprocal lattice of a two-dimensional crystal. Draw the first Brillouin zone into this reciprocal lattice.

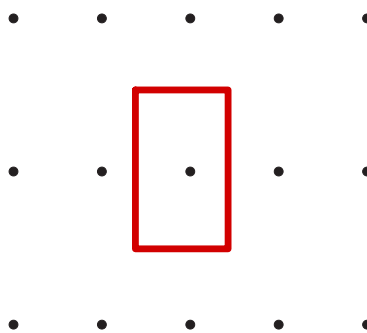


Figure 2: A two-dimensional reciprocal lattice.

3 Lattice waves (1 point)

Consider a Bravais lattice $\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2 + o\mathbf{a}_3$ and the corresponding reciprocal lattice $\mathbf{G} = m'\mathbf{b}_1 + n'\mathbf{b}_2 + o'\mathbf{b}_3$. Give a general expression for a plane wave with the periodicity of the Bravais lattice.

$$e^{i\mathbf{G} \cdot \mathbf{r}}$$

4 Vibrations (4 points)

Figure 3 shows the vibrational spectrum and different vibrational modes for a chain of with two atoms per unit cell and lattice constant b .

- Assign the vibrational patterns shown on the right hand side to the correct points on the dispersion curves on the left hand side. You can do this by drawing arrows on the figure pointing to the correct branch and the approximate value of k . Assume that the bright atoms are much lighter than the dark ones (2 points).
- Estimate the speed of sound in the chain. In order to do this, let the lattice constant be $3.14 \text{ \AA}$. Give an approximate value in m/s. (2 points).

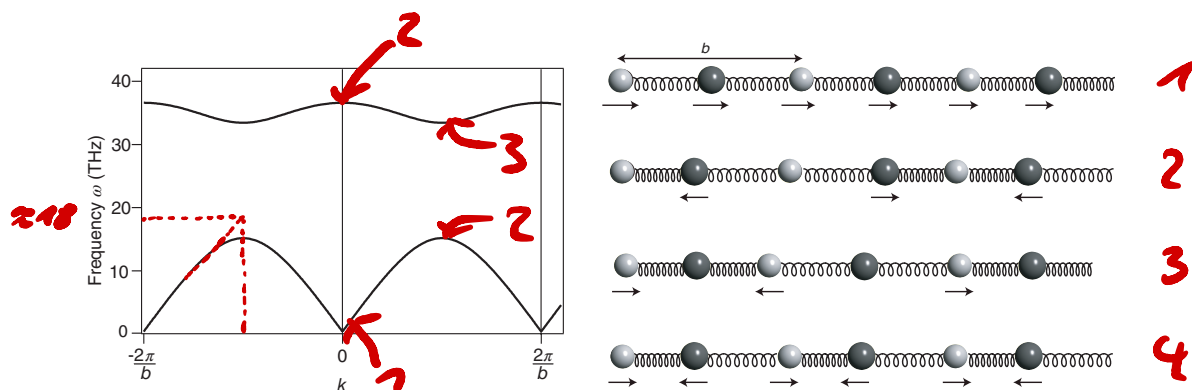


Figure 3: A two-dimensional reciprocal lattice.

$$v = \frac{\omega}{k} \approx \frac{\omega(\pi/b)}{\pi/b} = \frac{3.14 \text{ \AA} \cdot 18 \text{ THz}}{\pi} = 10^{-10} \text{ m}$$

5 Heat capacity (1 point)

For low temperatures, the heat capacity of a solid is found to be well-described by

$$C = \frac{\pi^2}{3} k_B^2 T g(E_F) + \frac{12\pi^4}{5} N k_B \left(\frac{T}{\Theta_D} \right)^3. \quad (1)$$

Is the solid an insulator or a metal? Give a brief explanation why.

metal because of first term

$$18 \cdot 10^{12} \text{ Hz} = 1800 \text{ m s}^{-1}$$

6 Density of states (2 points)

Using the concept of the density of states, give an approximate expression (i.e. a short formula) for

- the number of electrons contributing to the heat capacity of a metal (1 point). $g(E_F) kT$
- the number of electrons contributing to Pauli paramagnetism in a metal (1 point). $g(E_F) \cdot \mu_B \cdot \delta_0 / 2$

7 Band structure I (2 points)

Figure 4 shows the Brillouin zone and the band structure of a semiconductor. The horizontal axis represents different high-symmetry lines through the Brillouin zone.

- Indicate the location of the absolute band gap. (1 point)
- How many valence electrons must each unit cell of this material at least contain? (1 point)

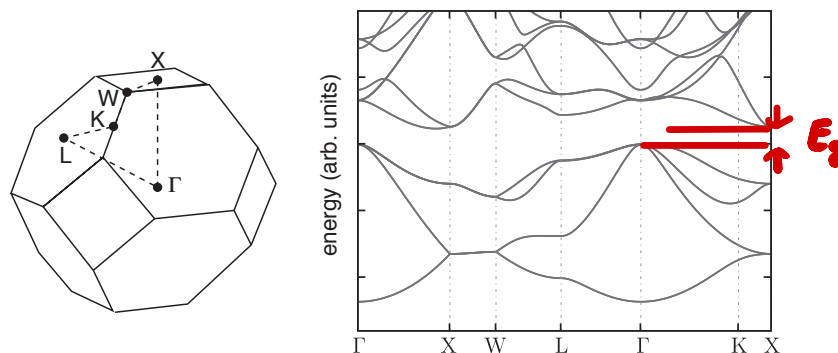


Figure 4: Brillouin zone and band structure of a semiconductor.

4 Filled bands $\Rightarrow 8e^-$ (more if there are more degenerate bands)

8 Band structure II (1 point)

Figure 5 shows the tight-binding band structure for an s orbital in an fcc lattice. The bands are plotted along high symmetry lines connecting the Γ , X and L points. Write the three high symmetry points on the horizontal axis at the appropriate positions. Hint: The coordinates of these high symmetry points are: $X = (2\pi/a)(1, 0, 0)$; $\Gamma = (0, 0, 0)$; $L = (2\pi/a)(1/2, 1/2, 1/2)$, where a is the lattice constant. The Brillouin zone is the same as for the semiconductor in Figure 4.

9 Effective mass (1 point)

The dispersion of an electronic band shall be given by $E(k) = A + Bk^2 + Ck^4$, where A , B and C are constants. Calculate the effective mass at $k = 0$.

$$m^* = \hbar^2 \left(\frac{d^2 E(k)}{dk^2} \right)^{-1} \bigg|_{k=0} = \frac{\hbar^2}{2B}$$

$$\frac{d^2 E}{dk^2} = 2B + 12Ck^2$$

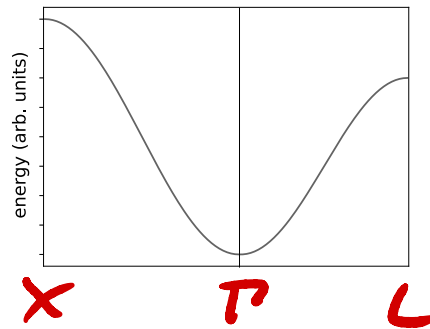


Figure 5: Brillouin zone and band structure of a semiconductor.

10 Magnetism (1 point)

How does the magnetic susceptibility of a paramagnetic insulator depend on the temperature? Draw your result into Figure 6.

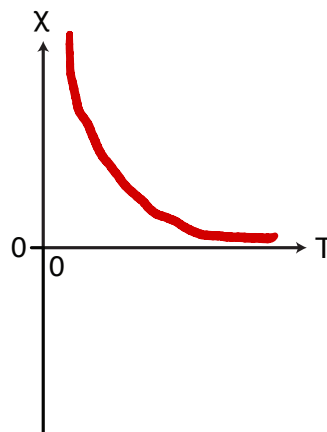


Figure 6: Temperature-dependent susceptibility of a paramagnetic material.

11 Superconductivity (1 point)

In a superconducting metal at $T = 0$ K, what fraction of the conduction electrons is bound in Cooper pairs (give an expression or state the order of magnitude)?

$$\Delta \cdot g(E_F)$$

Answers

Problem 1

a)

$$Z = (e^{-\beta\epsilon} + e^{\beta\epsilon})(e^{-\beta\epsilon'} + 1 + e^{\beta\epsilon'}), \quad \beta = \frac{1}{kT}.$$

b)

$$\bar{E} = \frac{e^{-\beta\epsilon} - e^{\beta\epsilon}}{e^{-\beta\epsilon} + e^{\beta\epsilon}}\epsilon + \frac{e^{-\beta\epsilon'} - e^{\beta\epsilon'}}{e^{-\beta\epsilon'} + 1 + e^{\beta\epsilon'}}\epsilon'$$

$$\lim_{T \rightarrow 0} \bar{E} = \lim_{T \rightarrow 0} \frac{e^{-2\beta\epsilon} - 1}{e^{-2\beta\epsilon} + 1}\epsilon + \frac{e^{-2\beta\epsilon'} - 1}{e^{-2\beta\epsilon'} + e^{-\beta\epsilon'} + 1}\epsilon' = -\epsilon - \epsilon'$$

The system is in the ground state at low temperature.

$$\lim_{T \rightarrow \infty} \bar{E} = \frac{1-1}{1+1}\epsilon + \frac{1-1}{1+1+1}\epsilon' = 0$$

All states are equally probable at high temperature, since this maximizes the entropy.

Problem 2

a)

$$S = k \ln \left(\frac{M(M-1)(M-2)}{6} \right)$$

b)

$$\bar{N} = \frac{M}{1 + e^{\beta(\eta-\mu)}}, \quad \beta = \frac{1}{kT}$$

Problem 3

a)

$$U = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} \frac{\epsilon_{n_x, n_y, n_z}}{e^{\beta(\epsilon_{n_x, n_y, n_z} - \mu)} - 1}, \quad \epsilon_{n_x, n_y, n_z} = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

b)

$$\mu = \frac{3h^2}{8mL^2} - \frac{kT}{N(1 - (T/T_c)^{3/2})}$$

c)

$$\begin{aligned} \left(\frac{\partial U}{\partial T}\right)_{V,N} = & \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} \frac{\epsilon_{n_x,n_y,n_z} e^{\beta(\epsilon_{n_x,n_y,n_z} - \epsilon_{1,1,1}) + \frac{1}{N(1-(T/T_c)^{3/2})}}}{\left(e^{\beta(\epsilon_{n_x,n_y,n_z} - \epsilon_{1,1,1}) + \frac{1}{N(1-(T/T_c)^{3/2})}} - 1 \right)^2} \\ & \times \left(\frac{\epsilon_{n_x,n_y,n_z} - \epsilon_{1,1,1}}{kT^2} - \frac{3T^{1/2}}{2NT_c^{3/2}(1-(T/T_c)^{3/2})^2} \right) \end{aligned}$$